

David Elliott Perryman

 website |  David Elliott Perryman |  elliottperryman |  elliottperryman@hey.com

SKILLS

Languages English, French(A2), C, C++, CUDA, Julia, Python, R, SQL
Methods Monte-Carlo, Distributed Optimization, Gaussian Processes, Digital Signal Processing

WORK

PhD Candidate at Institut Laue-Langevin March 2024 - present

- Developing autonomous experimentation methods for neutron scattering experiments on quantum materials.
- Supervisors: Martin Boehm at the Institut Laue Langevin, Daniel Mazzone at the Paul Scherrer Institute, and Marie-Bernadette at Université Grenoble Alpes.

Master Internships at Institut Laue-Langevin Summer 2022 & Spring 2023

- Writing symmetric Gaussian Processes for autonomous experimentation on triple axis spectroscopy.

Software Developer at Caravel Concepts Jan 2021 - Aug. 2021

- Wrote django+PostgreSQL database models and API endpoints
- Improved quality of investment plans by improving asset allocation optimization with JAX, Numba, and some basic math.

Undergraduate Researcher at Lawrence Berkeley Laboratory Aug. 2019 - Jun 2020

- Sped up ALS-steering software GPCAM by at least 20X through batching, converting Cholesky decomposition to linear solves, and porting to GPU with pytorch.
- Wrote fast numerical integration methods in Python for crystallography.
- Wrote distributed multi-optima optimization method.
- Wrote exploration scheme for material science application. Presented summary poster at SULI spring 2020.

Undergraduate Researcher at Oak Ridge National Labs Aug. 2017 - Jun 2022

- Ported signal processing convolution to CUDA FFT on GPU for 30X speedup
- Characterized and optimized signal feature extraction using Monte-Carlo methods
- Wrote signal anomaly detection with Conv Nets and data exploration with k-Means

Lumber tote at Columbia Construction and Crawlspace 2016 (Seasonal)

- Crawlspace Work
- Framing

Seasonal work in summer during high school.

Worker at Armstrong, Lacey, Shoup, and Williams Farms 2011 - 2016 (Seasonal)

- Landscaped
- Hauled Hay
- Farm upkeep
- Farming (tilling, pulling weeds, etc)

Seasonal work in summer during high school.

PROJECTS

- Pitched a business idea in the 2018 NAE Grand Challenge Pitch Competition (3 weeks of work)

- Reviewed articles for Pursuit, UTK’s undergraduate Journal (2017-2021)
- Developed Monte Carlo simulation for Helium microscopy (2018, about 4 months of work)
- Poured concrete accessibility ramps at Chickasaw park for an Eagle Scout Project (2015) (40 hours of work)

EDUCATION

PhD (Physics) at Université Grenoble Alpes	Mar 2024 - present
Master’s (Applied Math) at Institut Polytechnique de Grenoble	Aug 2021 - May 2023
Bachelor’s (Comp Sci & Physics) at UT Knoxville	Sept 2016 - May 2021

PUBLICATIONS

- [1] Francisco M Gonzalez et al. “First Full Dalitz Plot Measurement in Neutron β -Decay using the Nab Spectrometer and Implications for New Physics”. In: *arXiv preprint arXiv:2508.16045* (2025).
- [2] David Elliott Perryman et al. “Effect of likelihood misspecification in Gaussian process-driven autonomous experimentation”. In: *APL Machine Learning* 3.4 (2025).
- [3] Martin Boehm et al. “Autonomous Neutron Experiments”. In: *Methods and Applications of Autonomous Experimentation*. Chapman and Hall/CRC, 2023, pp. 256–275.
- [4] Marcus Noack et al. “GpCAM”. In: *Zenodo* (2022).
- [5] Marcus Noack and David Perryman. “fvGP v2. 2.0”. In: *US Department of Energy (DOE) Software* (2021), p. 26.
- [6] Marcus Michael Noack et al. “High-performance hybrid-global-deflated-local optimization with applications to active learning”. In: *2021 3rd Annual Workshop on Extreme-scale Experiment-in-the-Loop Computing (XLOOP)*. IEEE. 2021, pp. 24–29.
- [7] Marcus Noack and David Perryman. *HGDL v1*. Tech. rep. Lawrence Berkeley National Laboratory (LBNL), Berkeley, CA (United States), 2020.
- [8] Petrus H Zwart and Elliott D Perryman. “Evaluating crystallographic likelihood functions using numerical quadratures”. In: *Biological Crystallography* 76.8 (2020), pp. 736–750.
- [9] Jason Fry et al. “The Nab experiment: A precision measurement of unpolarized neutron beta decay”. In: *EPJ Web of Conferences*. Vol. 219. EDP Sciences. 2019, p. 04002.
- [10] David Perryman, Nab Experiment Collaboration, et al. “(CEU) Pile-up Detection in Silicon Detector Signals via Machine Learning for the Nab Experiment”. In: *APS Division of Nuclear Physics Meeting Abstracts*. Vol. 2019. 2019, SL-006.
- [11] Tom Shelton et al. “A GPU Algorithm to Determine the Time-of-Flight of Protons in the Neutron Decay Experiment Nab”. In: *APS April Meeting Abstracts*. Vol. 2019. 2019, S01-060.
- [12] David Perryman. “Optimization of Particle Energy and Arrival Time Determination using GPUs for the Nab Experiment”. In: *Bulletin of the American Physical Society* 63 (2018).

PRESENTATIONS

- Invited talk at Theoretical and Experimental Magnetism Meeting (accepted July 2026)

- Poster at International Conference on Neutron Scattering (accepted July 2025)
- Talk at College 2, Institut Laue-Langevin (January 2025)
- Talk at AI for X-ray and Neutron Scattering, Munich (April 2024)
- Invited 20 min talk at LBL Autonomous Experimentation Workshop (April 2021)
- Perryman, David (2018). “Optimization of Particle Energy and Arrival Time Determination using GPUs for the Nab Experiment”. In: Bulletin of the American Physical Society 63.
- Perryman, David, Nab Experiment Collaboration, et al. (2019). “(CEU) Pile-up Detection in Silicon Detector Signals via Machine Learning for the Nab Experiment”. In: APS Division of Nuclear Physics Meeting Abstracts. Vol. 2019, SL-006.
- Regional physics conference (SESAPS 2018) - won best oral presentation
- National Conference for Undergraduate Research (2019)
- UTK Eureka Symposium (Spring 2019)
- UTK Research Day (Spring 2019)